

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 27]

This question asks you to explore self-composite linear functions, of the form $f(x) = mx + c$, for varying values of m .

A function composed with itself is called a self-composite function.

For a function f , the function composition with itself is given by $(f \circ f)(x) = f(f(x))$.

Let f^n denote the n th composition of f with itself where $f^n(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{n \text{ times}}(x)$.

Hence, for example, $f^2(x) = (f \circ f)(x)$ and $f^3(x) = (f \circ f \circ f)(x) = f(f(f(x)))$.

Consider the linear function $f(x) = mx + c$, where $x \in \mathbb{R}$ and $m, c \in \mathbb{R}$.

(a) Show that

$$(i) \quad f^2(x) = m^2x + c(1 + m); \quad [3]$$

$$(ii) \quad f^3(x) = m^3x + c(1 + m + m^2). \quad [2]$$

(b) (i) Write down an expression for $f^4(x)$. [1]

(ii) Suggest a similar expression for $f^n(x)$, $n \in \mathbb{Z}^+$. [2]

(iii) By using your expression from part (b)(ii), or otherwise, find an expression in terms of n for $f^n(x)$ when $m = 1$. [3]

(c) For $m \neq 1$, use mathematical induction to prove that

$$f^n(x) = m^n x + c \left(\frac{1 - m^n}{1 - m} \right) \text{ for } n \in \mathbb{Z}^+. \quad [8]$$

(This question continues on the following page)

(Question 1 continued)

Consider $f^n(x) = m^n x + c \left(\frac{1 - m^n}{1 - m} \right)$ where $-1 < m < 1$.

- (d) As $n \rightarrow \infty$, the family of graphs $y = f^n(x)$ approaches the graph of a straight line, L . Determine the equation of L , giving your answer in terms of c and m . [4]

Consider $f^n(x) = m^n x + c \left(\frac{1 - m^n}{1 - m} \right)$ where $m = -1$.

- (e) (i) Show that $f^n(x) = -x + c$ when n is odd. [2]
- (ii) Find an expression for $f^n(x)$ when n is even. [2]